Ashvin Anilkumar

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Summary

Graduate ECE student at UW-Madison specializing in **Embodied AI** and **Sim-to-Real Transfer**. Expert in developing end-to-end robotics applications and integrating complex hardware workcells using **C++, Python**, and **ROS 2**. Experienced in **multi-agent reinforcement learning** and the creation of intuitive developer interfaces/APIs for robot simulation and control.

Education

University of Wisconsin-Madison, Master of Science in Electrical and Computer Engineering — December 2026

- *Relevant Coursework:* Computer Vision, Topics in Advanced Robotics, High-performance Computing, Optimization.

Government Engineering College, Barton Hill, Trivandrum, Bachelor of Technology in Electrical and Electronics Engineering — July 2023

- *Additional Coursework:* Minor in Information Technology

Research & Academic Projects

Multi-Agent Distributed Coordination(RoboCup) — December 2025 – Present

Isaac Sim | MuJoCo | ROS2 | Pytorch | Rerun

- Designing a squad of Booster T1s that could autonomously play soccer by implementing multi-agent robotic coordination using **ROS2** and **Deep RL**, under the guidance of Prof. Josiah Hanna.
- Supported in migrating from **Isaac Sim** to **MuJoCo** for training the robots in simulation, making the framework system-agnostic and improving the reliability of **Sim-to-Real transfer** by **40%**.
- Developing **Localization & Motion Planning** algorithms to optimize agent movement in unpredictable, dynamic sports environments.

University Rover Challenge: Perception & Autonomous Navigation — Sep 2025 – Dec 2025

ROS2 | Gazebo | DepthAI | RViz| RANSAC

- Programmed a real-time model-less **perception and mapping** framework for autonomous navigation on unstructured terrain using **ROS2 Humble** and **DepthAI**, optimized for low-latency inference on **Nvidia Jetson Nano**.
- Developed a ground detection algorithm using **RANSAC** and improved the system reliability by **30%**.
- Implemented **Path Planning** and **Movement** controls using ROS2, integrating stereo vision (DepthAI) and IMU data for sensor fusion.

Work Experience

Software Engineer, Backend Development(AI/ML), Qburst Technologies - Trivandrum, India — Aug 2023 – Aug 2025

Scalable AI Search & Distributed Systems

FastAPI | Langchain | Llamaindex | Postgres | MongoDB | Huggingface

- Deployed a **cloud-native microservices** using FastAPI and Python to handle complex global influence data.
- Implemented **agentic workflows** (RAG), optimizing token usage by **56%** and enhancing system response speed.
- Increased the accuracy of search results by **90%** and expanded the range of questions it can handle by **85%**.

AI Report Reviewer System

FastAPI | Langchain | Azure

- Formulated an AI system for automated evaluation and structured feedback on reports using **Langchain** and **FastAPI**.
- Achieved **89%** alignment with human ratings, improved evaluation accuracy by **35%** and reduced review costs by **67%**.

Technical Skills

Languages: Python, C++, Rust, JavaScript

Simulation & Tooling: Isaac Sim, MuJoCo, Gazebo, OpenAI Gym, NVIDIA Omniverse

ML & Data: PyTorch, TensorFlow, HuggingFace, LangChain, LlamaIndex

Robotics & Perception: ROS2(Humble), SLAM, OpenCV, Sensor Fusion

Systems & Performance: Linux, Git, Docker, Kubernetes, CUDA, OpenMPI

Databases: PostgreSQL, MongoDB, PGVector

Parallel Computing: CUDA

Hardware: Raspberry Pi, STM32, Jetson Nano

PCB Designing: Altium, KiCad

Open Source Contributions: Langchain, LlamaIndex, Visual Studio Code